18 October 2026

Editor-in-Chief
Journal of Statistical Computation and Simulation

Dear Editor,

I am pleased to submit the manuscript "How Stable Are Regression Conclusions Across Subpopulations? A Finite-Sample Empirical Study" for consideration in the *Journal of Statistical Computation and Simulation*.

The manuscript examines when a single pooled linear regression remains an adequate summary across observed subpopulations. It proposes a set of descriptive diagnostics covering coefficient signs, standardized coefficient drift, covariate-context adjustment, covariate shift, and the within-group cost of using the pooled rather than a group-specific model. A unified simulation study uses four data-generating scenarios, 140 settings per scenario, and 200 Monte Carlo replications per setting. The paper also benchmarks the diagnostics against the conventional group-slope interaction test and includes repeated held-out validation in an age-stratified empirical illustration.

The work fits the journal's scope in computational statistics and simulation. It provides a reproducible simulation design and practical diagnostic quantities for analysts who must decide whether pooling is defensible. The interaction benchmark clarifies the relationship between the proposed diagnostics and standard hypothesis testing: the diagnostics are intended to complement, not replace, interaction tests.

This manuscript is original, has not been published, and is not under consideration by another journal. Two versions are provided as requested: one with author details and one anonymized for peer review. The author declares no competing interests. No specific funding supported this research. Generative AI tools were used for code development, figure-generation workflows, and language editing, as stated in the manuscript; the author directed the research, reviewed the analysis, and takes responsibility for all claims.

Sincerely,

Nadjib Tebbal
American University of Beirut, Beirut, Lebanon
ORCID: 0009-0001-0462-1766
Email: as registered in the submission system